



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)

Re-Accredited by NAAC with 'A++' Grade

ANISH NAIR

PRN: 25070122037

BATCH: A2

ASSIGNMENT No: 4

TITLE: Develop a Java application demonstrating the use of wrapper classes for conversion between primitive data types and objects, and performing basic operations on wrapped values.

Problem Statement

Develop a Java application demonstrating the use of wrapper classes for conversion between primitive data types and objects, and performing basic operations on wrapped values.

Learning Objectives

To perform conversion between primitive data types and wrapper objects using Java wrapper classes.

Theory

Wrapper classes are predefined classes that encapsulate primitive data types into objects. Java provides wrapper classes such as Integer, Double, Character, and Boolean for every primitive type. They support autoboxing, unboxing, parsing, and conversion methods. Wrapper classes are widely used in Java Collections Framework and when object representation of primitive values is required.

Outcome

VS code Folder stuff > J as4.java

```
1  public class as4 {
2      String name;
3      int rollNo;
4
5      // Default constructor
6      as4() {
7          name = "Unknown";
8          rollNo = 0;
9      }
10
11     // Parameterized constructor
12     as4(String n, int r) {
13         name = n;
14         rollNo = r;
15     }
16
17     // Copy constructor
18     as4(as4 s) {
19         name = s.name;
20         rollNo = s.rollNo;
21     }
22
23     void display() {
24         System.out.println("Name: " + name + ", Roll No: " + rollNo);
25     }
26
27     public static void main(String[] args) {
28         as4 s1 = new as4();           // default
29         as4 s2 = new as4("Anish", 21); // parameterized
30         as4 s3 = new as4(s2);         // copy of s2
31
32         s1.display();
33         s2.display();
34         s3.display();
35     }
36 }
```

```
Name: Unknown, Roll No: 0
Name: Anish, Roll No: 21
Name: Anish, Roll No: 21
```

Conclusion: This experiment demonstrated how default, parameterized, and copy constructors simplify object initialization and data duplication in Java.

Exercise

1. Create a program to convert student marks from String format to Integer and calculate total marks.
2. Develop an **Employee Payroll System** that accepts employee IDs, basic salary, and bonus amounts from the user. Convert the entered values into wrapper objects and perform validation operations to ensure valid salary values before calculating the net salary.